GAURAV UPADHYAY

Software Engineer | Cloud & DevOps Engineer
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OBJECTIVE

Software Engineer with **3.5+ years of experience** building and deploying scalable web applications, with hands-on experience in **AWS, Docker, CI/CD, Linux, infrastructure automation, and cloud-native deployment practices**. Experienced in designing automated build and deployment pipelines, containerizing applications, managing cloud infrastructure, securing application configuration and secrets, and improving deployment reliability. Strong software engineering foundation across **Node.js, React, TypeScript and REST APIs**, enabling effective collaboration between application development and infrastructure teams.

WORK EXPERIENCE

R and P Innovations

July 2025 - Present
Jodhpur, RJ

Software Engineer

- Designed, developed, and deployed full-stack surveillance platforms processing large volumes of camera-generated vehicle detection data, with **200K+ detection images per day stored in Amazon S3 during high-volume weeks**.
- Built and maintained backend services using **Node.js, Express.js, and PostgreSQL**, deployed as containerized workloads on managed cloud servers with separate application and database environments.
- Designed and managed **AWS infrastructure using Terraform**, applying Infrastructure as Code practices to provision and maintain repeatable cloud environments.
- Implemented and maintained **CI/CD pipelines using Jenkins and GitHub Actions**, automating application builds, Docker image creation, testing, and deployment workflows.
- Implemented application monitoring and troubleshooting using **Amazon CloudWatch**, working with logs and operational metrics to identify and resolve deployment and runtime issues.
- Worked with AWS services including **EC2, S3, RDS, ECR, ECS, Lambda, CloudWatch, IAM, Route 53, and VPC**, supporting application deployment and cloud operations.
- Developed and maintained scalable frontend applications using **React.js, Next.js, and TypeScript**, giving the role a strong software-engineering foundation alongside cloud and platform responsibilities.

AIAB Ventures Pvt Ltd

March 2023 - March 2025
Mumbai, MH

Software Engineer

- Led development and scaling of full-stack applications using **React.js, Node.js, and MongoDB** to support multiple client-facing products.
- Designed and optimized **RESTful APIs for multi-database architectures**, enhancing backend performance, scalability, and data reliability.
- Built responsive, data-driven dashboards using **React.js and Vue.js**, boosting UI responsiveness and user engagement.
- Refactored legacy modules to reduce technical debt, improve performance, and streamline future development.

TECHNICAL SKILLS

- **Cloud & AWS:** AWS, EC2, VPC, S3, RDS, IAM, ECR, ECS, Lambda, CloudWatch, Route 53, ELB
- **DevOps & CI/CD:** Jenkins, GitHub Actions, CI/CD, Docker, Docker Compose, Git, GitHub
- **Infrastructure as Code:** Terraform
- **Containers & Orchestration:** Docker, Kubernetes, Helm, Amazon EKS
- **Monitoring & Observability:** Amazon CloudWatch, Prometheus, Grafana
- **Networking & Servers:** Linux, Nginx, SSH, VPC, Subnets, Security Groups, NACL, Bastion Hosts, DNS
- **Programming:** JavaScript, TypeScript, Python, Node.js, Bash

- **Backend & Databases**: Express.js, REST APIs, PostgreSQL, MongoDB, Redis
- **Frontend**: React.js, Next.js, Tailwind CSS, ShadCN
- **Architecture**: Three-Tier Architecture, Microservices, Containerized Applications, Cloud Deployment, Environment Management

EDUCATION

Masters in Computer Science *Aug 2021 - Aug 2023*

Lachoo Memorial College, Jodhpur Rajasthan

- Percentage: 74%

Bachelor in Computer Science *Jun 2018 - Jun 2021*

Lachoo Memorial College, Jodhpur Rajasthan

- Percentage: 75%

PROJECTS

Production-Style AWS Infrastructure with Terraform

- Designed and provisioned a secure AWS environment using **Terraform**, including VPC, public/private subnets, route tables, Internet/NAT connectivity, security groups, and a **bastion host**.
- Deployed a **three-tier application architecture** with separate frontend, backend, and database layers, following least-privilege networking and environment isolation practices.
- Managed infrastructure configuration as code to create reproducible **development and production environments**.

Containerized Application CI/CD Platform

- Containerized a full-stack application using **Docker and Docker Compose**, implementing reproducible multi-stage builds for application services.
- Built CI/CD pipelines using **Jenkins and GitHub Actions** to automate source checkout, testing, Docker image builds, and deployment workflows.
- Integrated **AWS ECR/ECS** for container image management and application deployment, with environment-specific configuration and secure secret handling.

Amazon EKS Deployment & Observability Platform (Personal)

- Deployed containerized applications on **Amazon EKS** using Kubernetes and **Helm**, implementing Kubernetes-native deployment and configuration practices.
- Configured Kubernetes resources including Deployments, Services, ConfigMaps, Secrets, probes and autoscaling components.
- Implemented an observability stack using **Prometheus and Grafana** to collect application/cluster metrics and build operational dashboards.
- Practiced production-oriented Kubernetes concepts including namespaces, resource management, rolling deployments, service discovery, and troubleshooting.

Cloud-Native Image Processing Platform

- Designed a cloud architecture for a high-volume vehicle detection workload using **Node.js, Express.js, PostgreSQL, Docker, and Amazon S3**.
- Implemented scalable object-storage workflows for camera-generated images, supporting workloads exceeding **200K+ vehicle detection images per day**.
- Used AWS services including **S3, Lambda, CloudWatch, IAM, Route 53, and compute/container services** to support application deployment, automation, monitoring, and operations.

CERTIFICATES

- DevOps Certification Course Udemy
- Web-Dev BootCamp Udemy